1-7

Roview of Mathematical Theory

Theory of Computation

Let us discuss each one by one-

i) Negation

horizontal bar on letter.

S	1s
T	F.
F	T

ii) Conjunction AND

truth table for this connective is

A	B	A^B 10
T	T	T.
T	F	F
F	T	F
F	F	F

ii)Disjunction OR

Jf A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T.	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denote byA-BorA = B.

Revlew of Mathematical Theory

1-8

Theory of Computation

The truth table for this connective is

A	8	A-B
T	T	T
T	F	F
F	T	T
F	F	T

For example: Find truth value for following proposition. If an apple is not green then it must be red" Let A denotes An apple is not green and B denotes It must be redThe statement A is false and B is true. Hence the proposition is true.

v) If and only if

If A and B are two propositions then A if and only if B.It can be denoted as AB The truth table will be

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For example: I will go to the school if and only if it is not rainingHere statement A is T will go to the school The statement B is It is not rainingThus the resultant proposition is false.

1.5 Relations

Relaonship is a major aspeci between two objects, even this is true in our real life. One oyect can be related with the other object by a some relation.Then those two objects forma pair based on this certain relationship. DelinitionThe relation R is a collction for the set S which represents the pair of elements.d

For example:(a.b is in R. We can represent their relation as a R b.The first pair is chosen from a set called range.

Reviow of Mathematical Theory

1-9

Theory of Computation

Properties of Relations

A relation R on set S is

1. Relexive if iRi for all i in S
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk.
4. Symmetric if iRj implies jRi.
5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive.

For example: ifA = ab| then

reflexive relation R can be=|a.a, b, b
irreflexive relation R can be =4.b)1
transitive relation R can be = a.b).b, aaa
symmetric relation R can be=(a.b)b, a|
asymmetric relation R can be =a.b1

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive over some set S. Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting to each other. A relation is said to be antisymmetric if a, b)e R and b, a)e R then a =a.

Example 1.5: Determine whether R is cquivalence relation or not where A = 0121R=/0, 0.1, 0.1, 1), 2, 2)2, 1

Solution: The R is reflexive because 0, 0).1, 1)2, 2e R. Where as R is not symmetric because (0, 1) is not in R whereas 1, 0) is in R. Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given, it may not be reflexive or transitive By adding some pairs we make the relation as reflexive or transitive.

For exampleLeta.b)bc)aa)bb